

SQL Learning Journal #6 – Aggregate Functions to Understand Data

Introduction

In today's lesson, I learned how to use SQL aggregate functions to summarize data. I practiced AVG(), SUM(), MIN(), MAX(), COUNT(), and ROUND() functions and learned how to combine them with filtering conditions.

1. Aggregate Functions

AVG(), SUM(), MIN(), MAX(), and COUNT() are aggregate functions used after SELECT. AVG() and SUM() work only with numerical values.

2. MIN(), MAX(), and COUNT()

MIN(), MAX(), and COUNT() work with numerical and text values. MIN() returns the first text alphabetically; MAX() returns the last.

3. Using Aliases

```
SELECT MAX(field_name) AS nick_name_max FROM table_name;
```

4. Combining Filtering with Aggregate Functions

```
SELECT AVG(year) AS average_year FROM table_name WHERE city='Istanbul';
```

```
SELECT SUM(year) AS sum_year FROM table_name WHERE city='Istanbul';
```

5. Using ROUND()

ROUND() rounds decimal values. Omitting the second argument is the same as using 0.

6. ROUND() with Positive and Negative Values

Positive values round digits to the right of the decimal point. Negative values round digits to the left.

What I Learned Today

I learned how to summarize data using aggregate functions and combine them with filtering conditions. Careful use of commas and parentheses is essential.